NEXUS

An autonomous, multimodal AI assistant with voice interaction, long-term memory, and OS/device automation — built module by module, each one fully tested before the next begins.

Status: Module 1 of N — Core Foundation ✅

See `MODULE_LOG.md` for the running architecture log across modules.

Quick start (Module 1)

```
# No third-party dependencies required for the core module itself.
python -m unittest discover -s tests -v
```

Expected: `Ran 32 tests ... OK`

To install dev tooling (pytest, mypy, ruff) for a nicer workflow:

```
pip install -r requirements.txt
pytest -v
```

Docker

```
docker build -t nexus:module-1 .
docker run --rm nexus:module-1
```

The image currently runs the test suite as its entrypoint (there is no
long-running service yet — that arrives with the agent orchestration and
voice pipeline modules).

Configuration

Copy `.env.example` to `.env` and adjust values. Every variable is documented there and validated in `nexus/core/config.py`.

Project layout so far

```
nexus/
    core/
        config.py       # Layered,
validated configuration (stdlib-only)
        logging.py      # Structured
JSON logging + console output
        di.py            #
Lightweight dependency injection
container
        interfaces.py    # Protocols:
LLMProvider, MemoryStore,
ActionExecutor, ...
        exceptions.py    # Exception
hierarchy
tests/
    test_config.py
    test_logging.py
    test_di.py
    test_interfaces.py
```